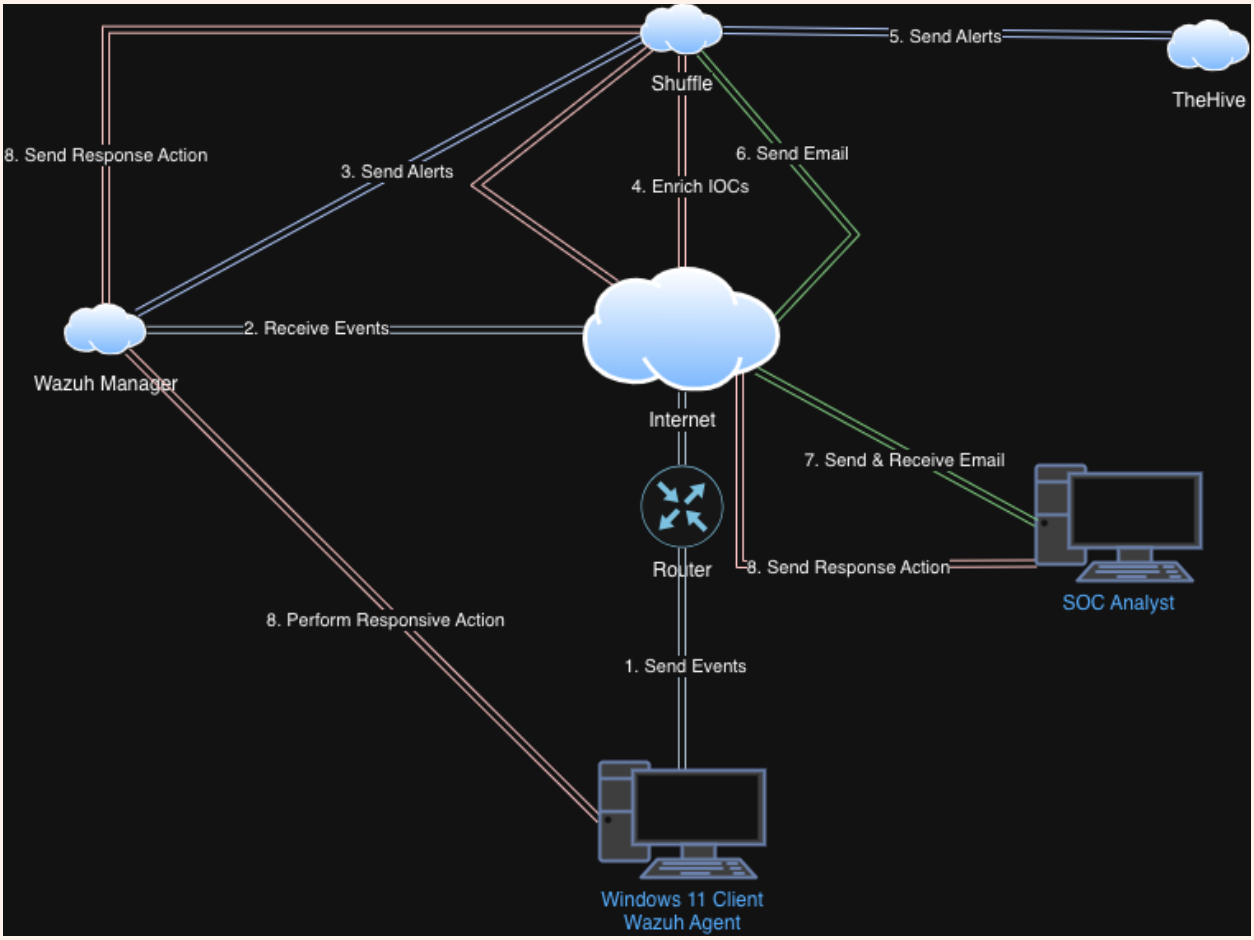
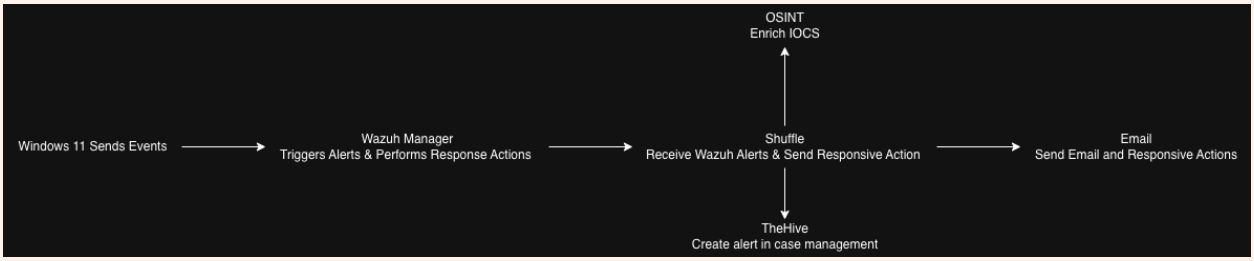
SOC Automation Project: Simulated Mimikatz Credential Dumping

High-Level Architecture Diagram:





Tools Used

VirtualBox VM – Client machine running Windows 10

Wazuh – Open Source XDR and SIEM capabilities in onw solution

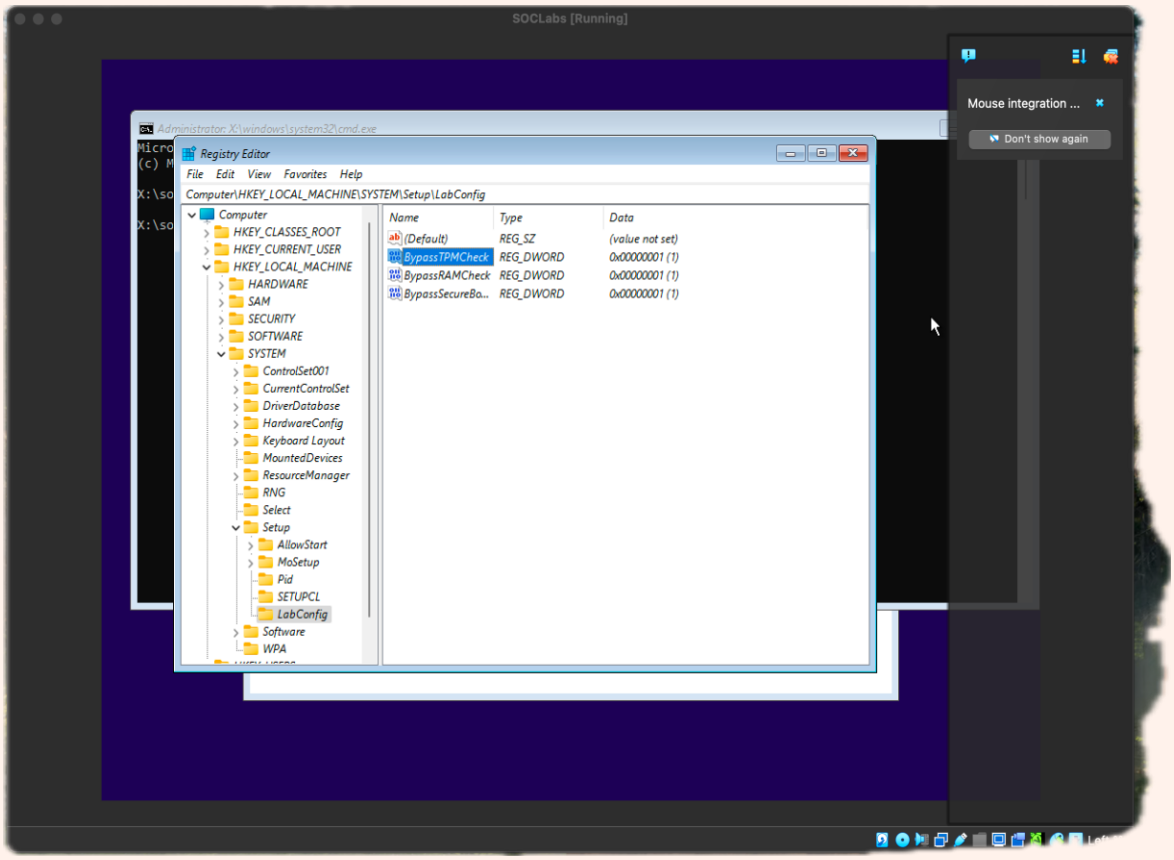
The Hive – Case Management System

Ubuntu 22.04 - Utilized for The Hive and Wazuh

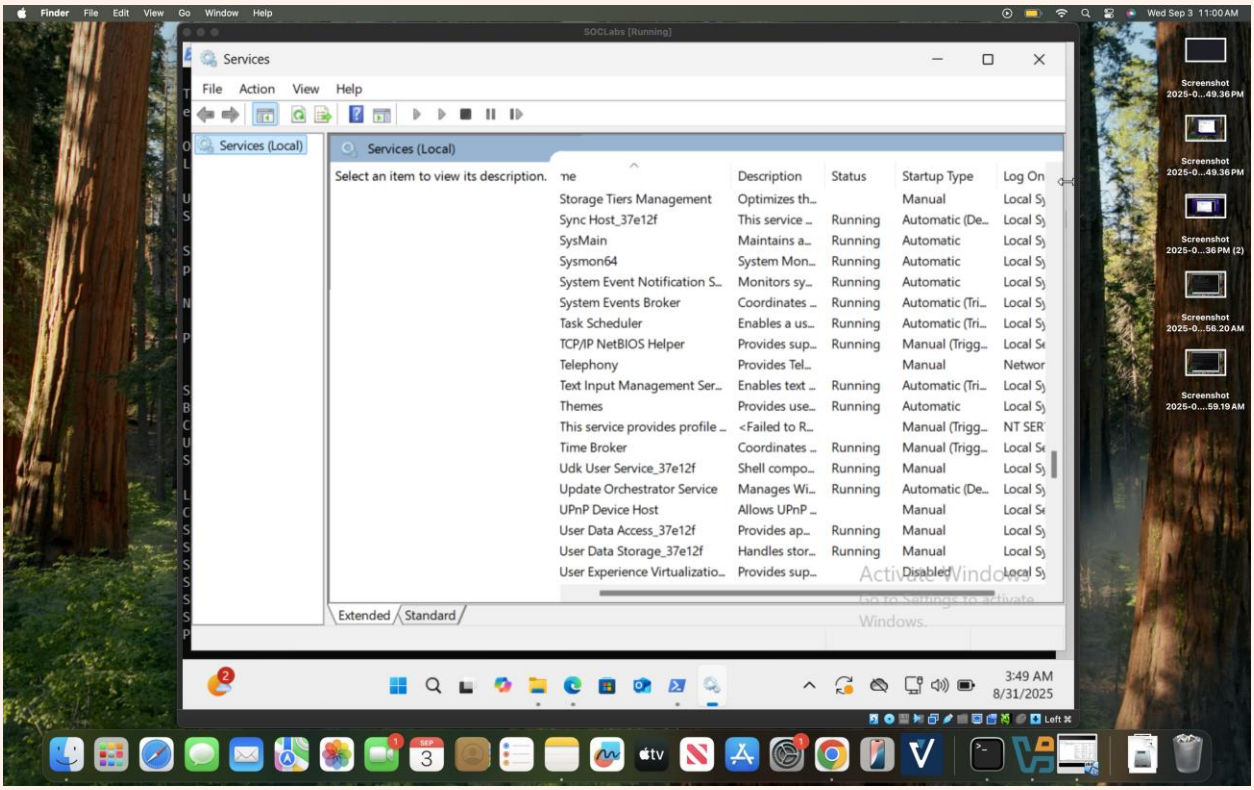
Sysmon (Enhanced Process Monitoring) - Sysmon logs detailed information about process creations, including the parent process, command line arguments, and hashes, allowing for better visibility into potentially malicious activities. Downloaded on the Windows 10 Client.

Shuffle (SOAR) – Used to create automated workflows by integrating webhooks together.

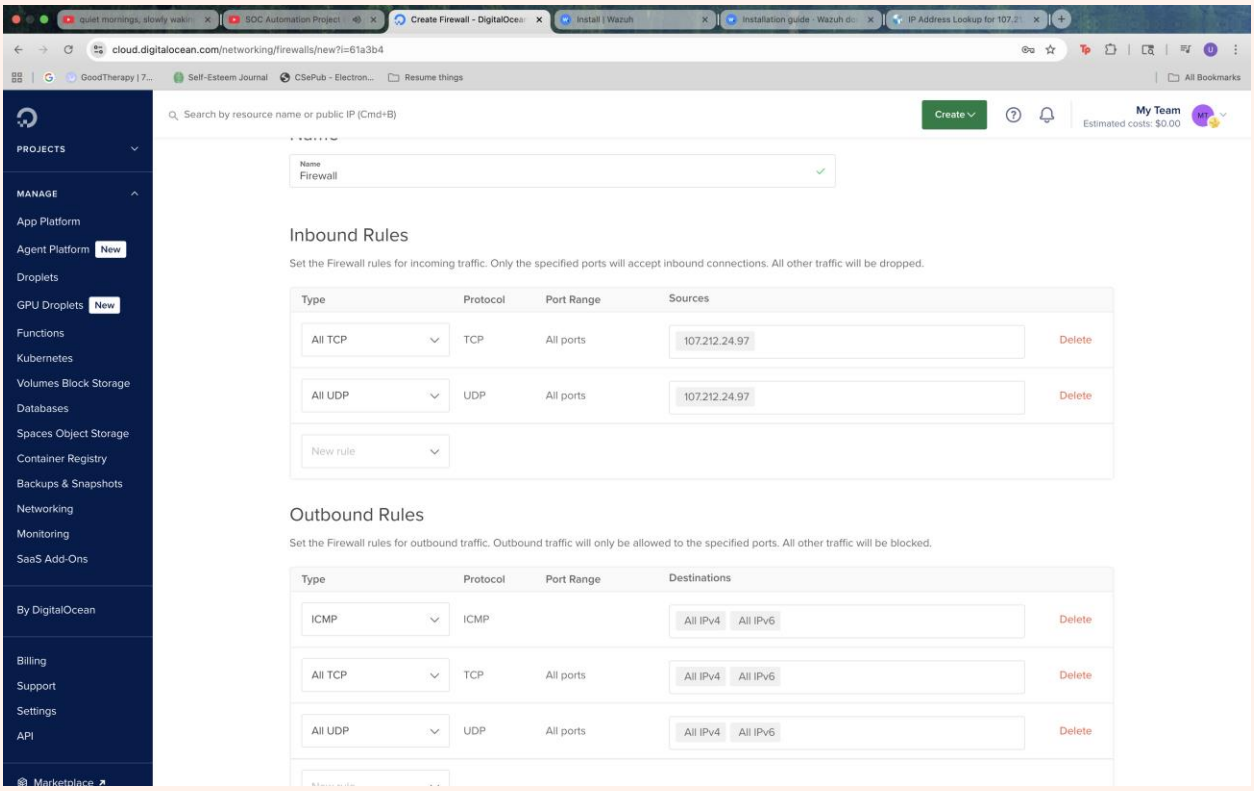
Description: Downloading windows 10 onto client. To bypass TPM and allow installation, configured setup.



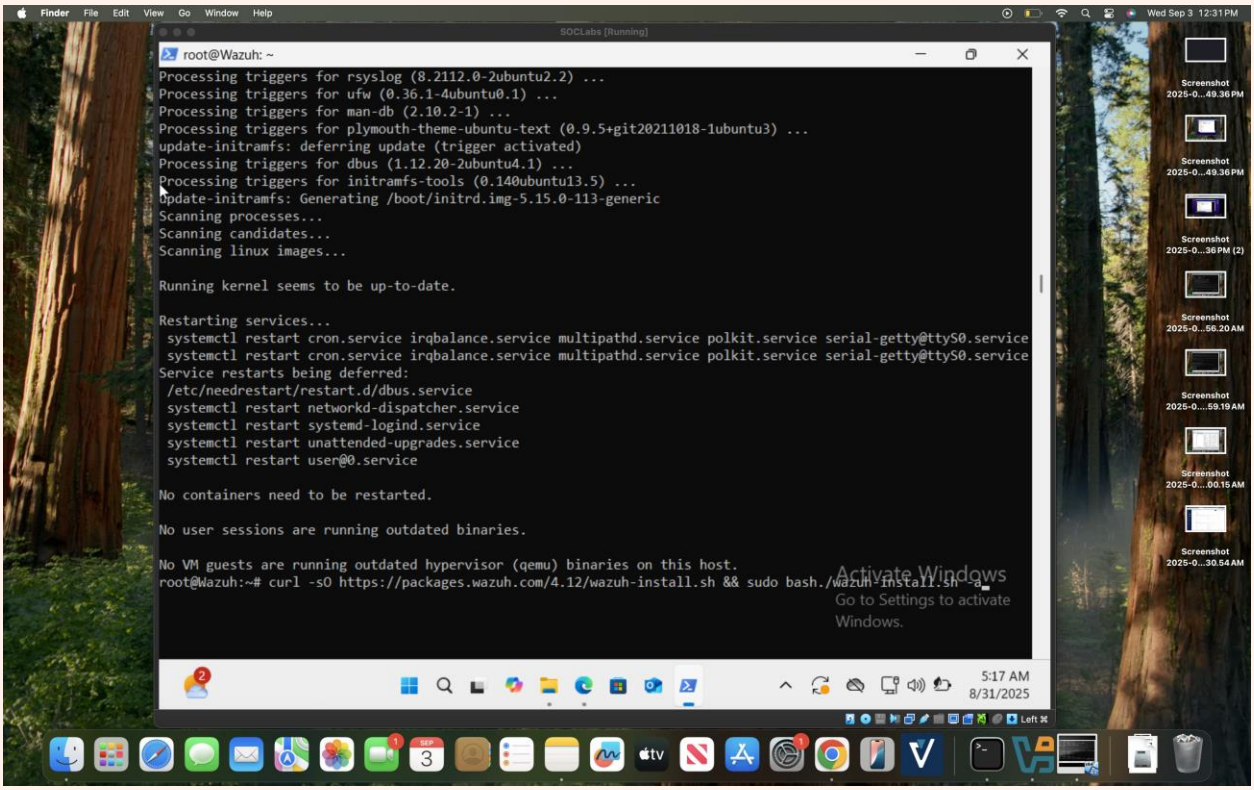
Description: Downloading Sysmon onto client and starting services.



Description: Creating Droplets on Digital Ocean and setting up firewall rules for both The Hive and Wazuh.



Description: Installing Wazuh and setting up for The Hive.



Description: Configuring Cassandra: changing listen, RPC address, and seed address to public IP of The Hive.

```
~ — root@thehive: ~ — ssh root@157.245.4.128        ~ — root@Wazuh: ~ — ssh root@104.131.168.175
GNU nano 6.2                                  /etc/cassandra/cassandra.yaml

# Cassandra storage config YAML

# NOTE:
#   See https://cassandra.apache.org/doc/latest/configuration/ for
#   full explanations of configuration directives
# /NOTE

# The name of the cluster. This is mainly used to prevent machines in
# one logical cluster from joining another.
cluster_name: 'MySoc'

# This defines the number of tokens randomly assigned to this node on the ring
# The more tokens, relative to other nodes, the larger the proportion of data
# that this node will store. You probably want all nodes to have the same number
# of tokens assuming they have equal hardware capability.
#
# If you leave this unspecified, Cassandra will use the default of 1 token for legacy compatibility,
# and will use the initial_token as described below.
#
# Specifying initial_token will override this setting on the node's initial start,
# on subsequent starts, this setting will apply even if initial token is set.
#
# See https://cassandra.apache.org/doc/latest/getting_started/production.html#tokens for
# best practice information about num_tokens.
#
num_tokens: 16

# Triggers automatic allocation of num_tokens tokens for this node. The allocation
# algorithm attempts to choose tokens in a way that optimizes replicated load over
# the nodes in the datacenter for the replica factor.
#
# The load assigned to each node will be close to proportional to its number of
# vnodes.
#
# Only supported with the Murmur3Partitioner.

# Replica factor is determined via the replication strategy used by the specified
# keyspace.
# allocate_tokens_for_keyspace: KEYSPACE

# Replica factor is explicitly set, regardless of keyspace or datacenter.
# This is the replica factor within the datacenter, like NTS.
allocate_tokens_for_local_replication_factor: 3

# initial_token allows you to specify tokens manually.  While you can use it with
# vnodes (num_tokens > 1, above) -- in which case you should provide a
# comma-separated list -- it's primarily used when adding nodes to legacy clusters
# that do not have vnodes enabled.
# initial_token:

# May either be "true" or "false" to enable globally
hinted_handoff_enabled: true

# When hinted_handoff_enabled is true, a black list of data centers that will not
# perform hinted handoff
# hinted_handoff_disabled_datacenters:
#    - DC1
#    - DC2

# this defines the maximum amount of time a dead host will have hints
# generated.  After it has been dead this long, new hints for it will not be
# created until it has been seen alive and gone down again.
# Min unit: ms
max_hint_window: 3h

# Maximum throttle in KiBs per second, per delivery thread.  This will be
# reduced proportionally to the number of nodes in the cluster.  (If there
# are two nodes in the cluster, each delivery thread will use the maximum
# rate; if there are three, each will throttle to half of the maximum,
# since we expect two nodes to be delivering hints simultaneously.)
# Min unit: KiB
hinted_handoff_throttle: 1024KiB
                                                [ Read 1877 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark  M-] To Bracket  M-Q Previous  ^B Back      ^Q Prev Word  ^A Home      ^P Prev Line
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy      ^O Where Was    M-W Next      ^F Forward   ^W Next Word  ^E End       ^N Next Line
```

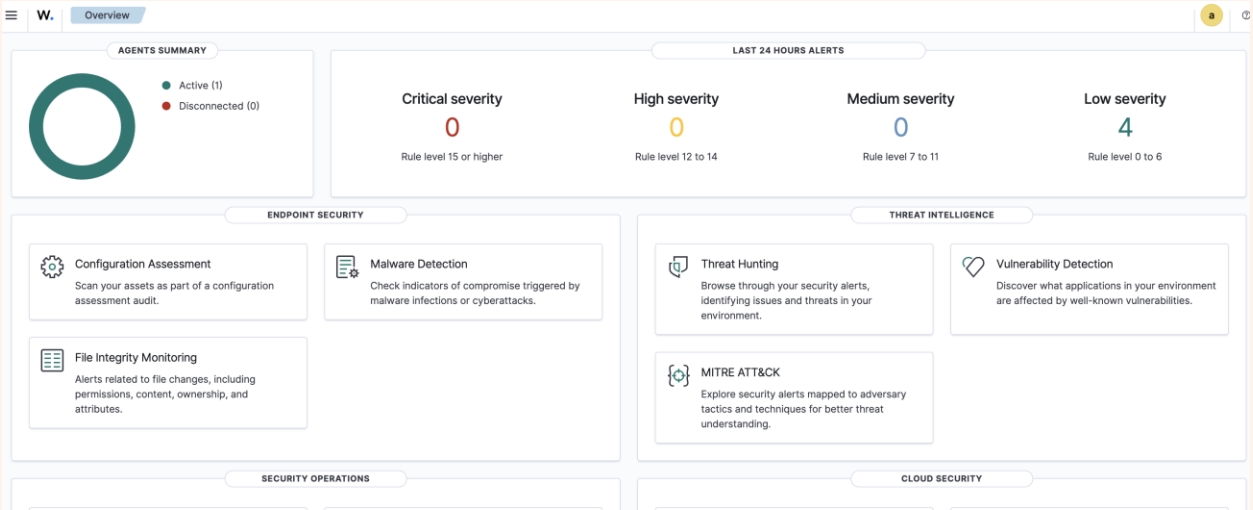
Description: Updating Elasticsearch configuration by changing the network host to the Hive IP address, uncommenting and configuring the HTTP port, setting cluster.initial_master_nodes to node-1, and defining discovery seed hosts to support cluster scaling. Starting Elasticsearch and verifying that the service is running and accessible.

```
  GNU nano 6.2
# ======================== Elasticsearch Configuration =========================
#
# NOTE: Elasticsearch comes with reasonable defaults for most settings.
#       Before you set out to tweak and tune the configuration, make sure you
#       understand what are you trying to accomplish and the consequences.
#
# The primary way of configuring a node is via this file. This template lists
# the most important settings you may want to configure for a production cluster.
#
# Please consult the documentation for further information on configuration options:
# https://www.elastic.co/guide/en/elasticsearch/reference/index.html
#
# ---------------------------------- Cluster -----------------------------------
#
# Use a descriptive name for your cluster:
#
cluster.name: thehive
#
# ------------------------------------ Node ------------------------------------
#
# Use a descriptive name for the node:
#
node.name: node-1
#
# Add custom attributes to the node:
#
#node.attr.rack: r1
#
# ---------------------------------- Paths -----------------------------------
#
# Path to directory where to store the data (separate multiple locations by comma):
#
path.data: /var/lib/elasticsearch
#
# Path to log files:
#
path.logs: /var/log/elasticsearch
#
# ---------------------------------- Memory ----------------------------------
#
# Lock the memory on startup:
#
#bootstrap.memory_lock: true
#
# Make sure that the heap size is set to about half the memory available
# on the system and that the owner of the process is allowed to use this
# limit.
#
# Elasticsearch performs poorly when the system is swapping the memory.
#
# ---------------------------------- Network ---------------------------------
#
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
#
network.host: 157.245.4.128
#
# By default Elasticsearch listens for HTTP traffic on the first free port it
# finds starting at 9200. Set a specific HTTP port here:
#
http.port: 9200
```

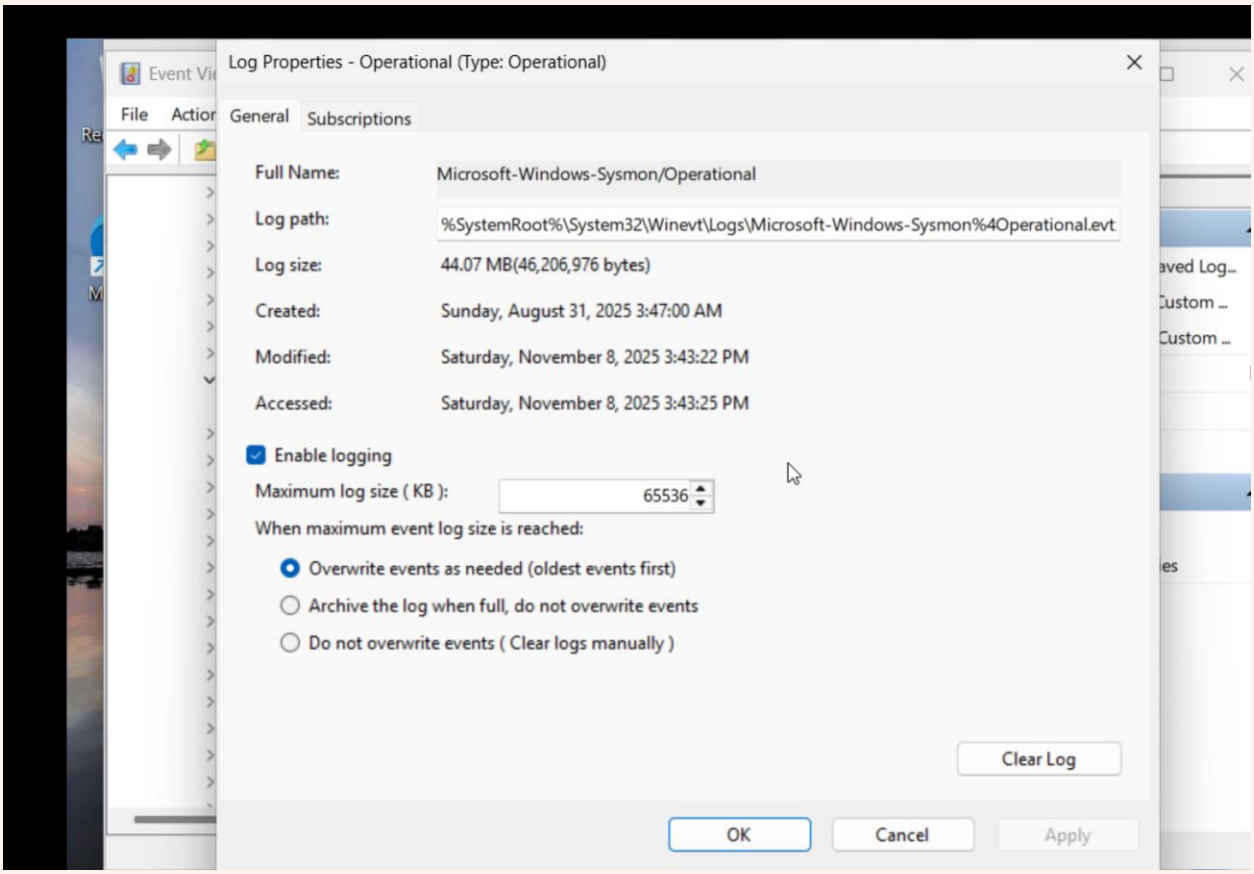
Description: Configuring TheHive by updating application.conf, changing the hostname to the Hive public IP address, updating the cluster name from Cassandra, setting the application base URL to the Hive public IP, and starting and enabling TheHive service.

```
  GNU nano 6.2
# TheHive configuration - application.conf
#
#
# This is the default configuration file.
# This is prepared to run with all services locally:
# - Cassandra for the database
# - Elasticsearch for index engine
# - File storage is local in /opt/thp/thehive/files
#
# If this is not your setup, please refer to the documentation at:
# https://docs.strangebee.com/thehive/
#
#
# Secret key - used by Play Framework
# If TheHive is installed with DEB/RPM package, this is automatically generated
# If TheHive is not installed from DEB or RPM packages run the following
# command before starting thehive:
#   cat > /etc/thehive/secret.conf << _EOF_
#   play.http.secret.key="$(cat /dev/urandom | tr -dc 'a-zA-Z0-9' | fold -w 64 |#   head -n 1)"
#   _EOF_
include "/etc/thehive/secret.conf"


# Database and index configuration
# By default, TheHive is configured to connect to local Cassandra 4.x and a
# local Elasticsearch services without authentication.
db.janusgraph {
  storage {
    backend = cql
    hostname = ["157.245.4.128"]
    # Cassandra authentication (if configured)
    # username = "thehive"
    # password = "password"
    cql {
      cluster-name = MySoc
      keyspace = thehive
    }
  }
  index.search {
    backend = elasticsearch
    hostname = ["157.245.4.128"]
    index-name = thehive
  }
}

# Attachment storage configuration
# By default, TheHive is configured to store files locally in the folder.
# The path can be updated and should belong to the user/group running thehive service. (by default: thehive:thehive)
storage {
  provider = localfs
  localfs.location = /opt/thp/thehive/files
}

# Define the maximum size for an attachment accepted by TheHive
play.http.parser.maxDiskBuffer = 1GB
# Define maximum size of http request (except attachment)
play.http.parser.maxMemoryBuffer = 10M

# Service configuration
application.baseUrl = "http://157.245.4.128:9000"
play.http.context = "/"

# Additional modules
#
# TheHive is strongly integrated with Cortex and MISP.
# Both modules are enabled by default. If not used, each one can be disabled by
# uncommenting the configuration line.
scalligraph.disabledModules += org.thp.thehive.connector.cortex.CortexModule
scalligraph.disabledModules += org.thp.thehive.connector.misp.MispModule
```

Description: Configuring Wazuh and integrating it with the virtual machine.



Description: Finding Sysmon event properties in Event Viewer to correctly map and name Event IDs in the OSSEC configuration file.

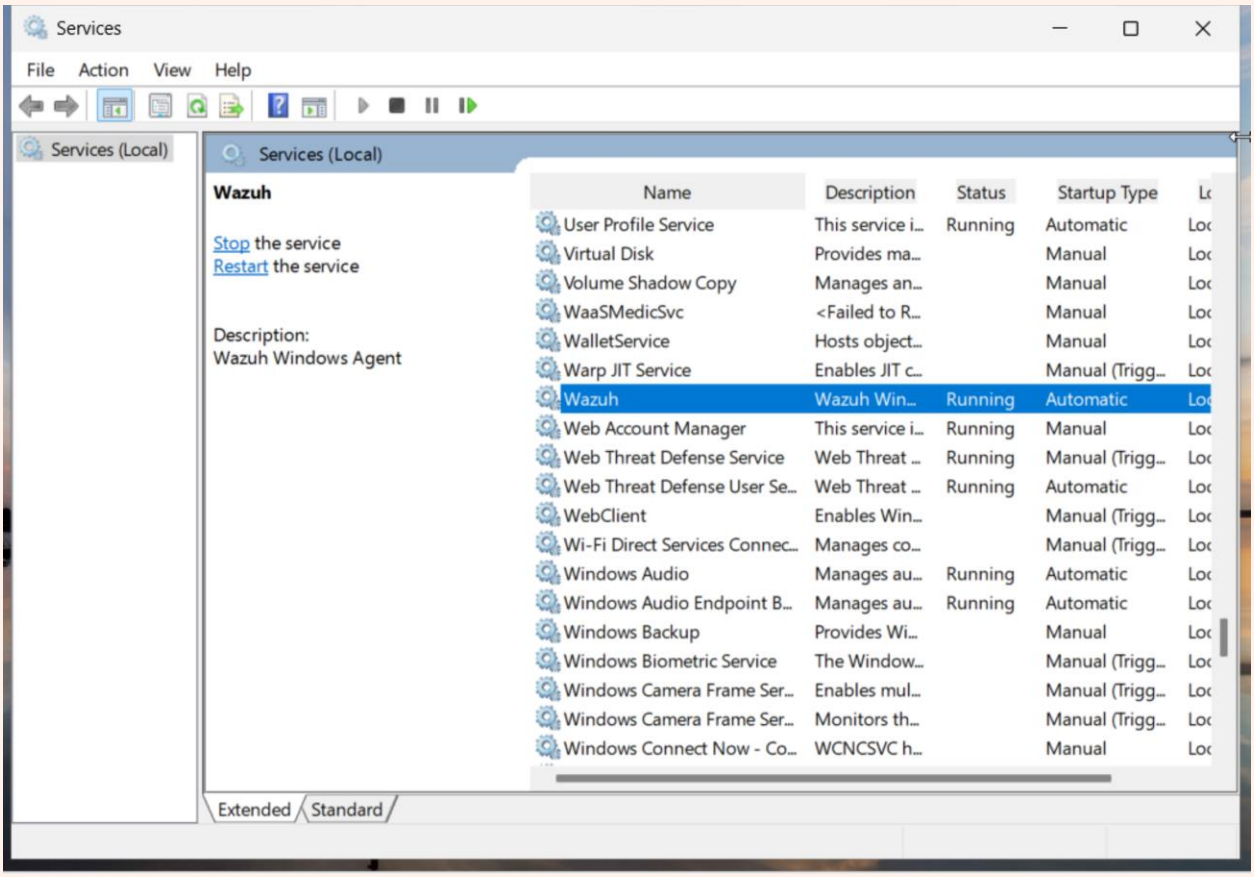


File Edit View

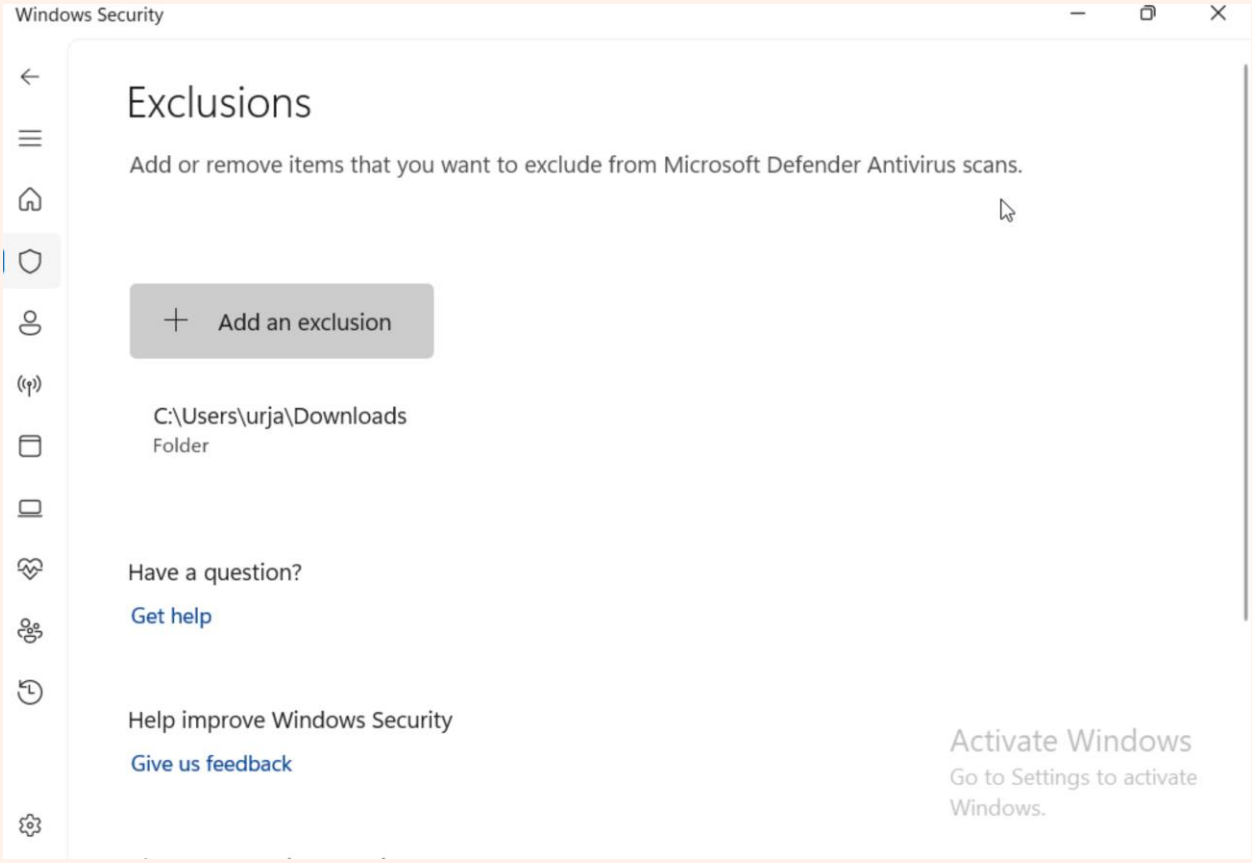
```
    <queue_size>5000</queue_size>
    <events_per_second>500</events_per_second>
  </client_buffer>

  <!-- Log analysis -->
  <localfile>
    <location>Microsoft-Windows-Sysmon/Operational</location>
    <log_format>eventchannel</log_format>
  </localfile>

  <localfile>
    <location>Application</location>
    <log_format>eventchannel</log_format>
  </localfile>

  <localfile>
    <location>active-response\active-responses.log</location>
    <log_format>syslog</log_format>
  </localfile>

  <!-- Policy monitoring -->
  <rootcheck>
    <disabled>no</disabled>
    <windows_apps>./shared/win_applications_rcl.txt</windows_apps>
    <windows_malware>./shared/win_malware_rcl.txt</windows_malware>
  </rootcheck>
```

Ln 1, Col 1 | 9,264 characters | Plain text | 100% | Windows (

Description: Restarting Wazuh services.



Description: Excluding downloads folder in exclusions on security settings of VM to be able to download Mimikatz.



Description: Configuring the Wazuh manager by updating the OSSEC configuration file, setting logall to yes, and restarting services. Enabling logall capabilities is allowing Wazuh to log all events under the archive.

```
GNU nano 6.2                                                    ossec.conf *
<!--
  Wazuh - Manager - Default configuration for ubuntu 22.04
  More info at: https://documentation.wazuh.com
  Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>yes</logall>
    <logall_json>yes</logall_json>
    <email_notification>no</email_notification>
    <smtp_server>smtp.example.wazuh.com</smtp_server>
    <email_from>wazuh@example.wazuh.com</email_from>
    <email_to>recipient@example.wazuh.com</email_to>
    <email_maxperhour>12</email_maxperhour>
    <email_log_source>alerts.log</email_log_source>
    <agents_disconnection_time>10m</agents_disconnection_time>
    <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
    <update_check>yes</update_check>
  </global>

  <alerts>
    <log_alert_level>3</log_alert_level>
    <email_alert_level>12</email_alert_level>
  </alerts>

  <!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
  <logging>
    <log_format>plain</log_format>
  </logging>

  <remote>
    <connection>secure</connection>
    <port>1514</port>
    <protocol>tcp</protocol>
    <queue_size>131072</queue_size>
  </remote>

  <!-- Policy monitoring -->
  <rootcheck>
    <disabled>no</disabled>
    <check_files>yes</check_files>
    <check_trojans>yes</check_trojans>
    <check_dev>yes</check_dev>
    <check_sys>yes</check_sys>
    <check_pids>yes</check_pids>
    <check_ports>yes</check_ports>
    <check_if>yes</check_if>

    <!-- Frequency that rootcheck is executed - every 12 hours -->
    <frequency>43200</frequency>

    <rootkit_files>etc/rootcheck/rootkit_files.txt</rootkit_files>
    <rootkit_trojans>etc/rootcheck/rootkit_trojans.txt</rootkit_trojans>

    <skip_nfs>yes</skip_nfs>

    <ignore>/var/lib/containerd</ignore>
    <ignore>/var/lib/docker/overlay2</ignore>
```

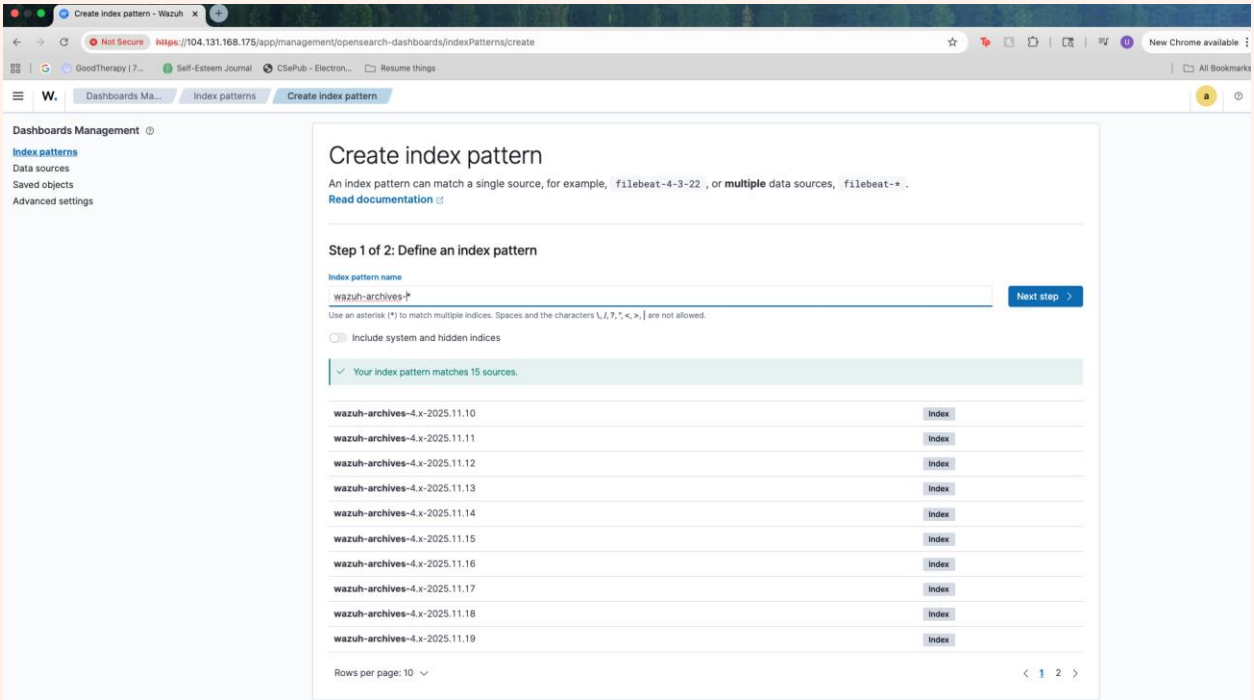
Description: Viewing archive.log in /var/ossec/logs/archives using cat to verify that Wazuh is logging events under the archive.

```
essed:\r\nRuleName: technique_id=T1036,technique_name=Masquerading\r\nUtcTime: 2025-11-09 02:50:46.924\r\nSourceProcessGUID: {fb82638e-d623-698f-6405-00000000700}\r\nSourceProcessId: 8504\r\nSourceThreadId: 9924\r\nSourceImage: C:\\Users\\urja\\AppData\
\\Local\\Microsoft\\OneDrive\\OneDrive.exe\r\nTargetProcessGUID: {fb82638e-f1a5-698f-6108-00000000700}\r\nTargetProcessId: 4800\r\nTargetImage: C:\\WINDOWS\\System32\\WindowsPowerShell\\\v1.0\\powershell.exe\r\nGrantedAccess: 0x101411\r\nCallTrace: C:\\WI
NDOWS\\SYSTEM32\\ntdll.dll+162d74|C:\\WINDOWS\\System32\\KERNELBASE.dll+c54f6|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncClient.dll+69e811|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSy
ncClient.dll+f22e8|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncClient.dll+f28ad|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncClient.dll+f1f99|C:\\Users\\urja\\AppData\\Local\\Microsof
t\\OneDrive\\25.194.1005.0003\\FileSyncEvents.dll+f0e0|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncHost.DLL+dc0f|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncHost.DLL+10485|C:\\WINDOW
S\\System32\\ucrtbase.dll+37b0|C:\\WINDOWS\\System32\\KERNEL32.DLL+2e8d7|C:\\WINDOWS\\SYSTEM32\\ntdll.dll+3c34c\r\nSourceUser: DESKTOP-437I1E6\\urja\r\nTargetUser: DESKTOP-437I1E6\\urja\"}","eventdata":{"ruleName":"technique_id=T1036,technique_name=Masqu
erading","utcTime":"2025-11-09 02:50:46.924","sourceProcessGUID":{fb82638e-d623-698f-6405-00000000700}","sourceProcessId":"8504","sourceThreadId":"9924","sourceImage":"C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\OneDrive.exe","t
argetProcessGUID":"{fb82638e-f1a5-698f-6108-00000000700}","targetProcessId":"4800","targetImage":"C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe","grantedAccess":"0x101411","callTrace":"C:\\\\WINDOWS\\\\SYSTEM32\\\\ntdll.dll+16
2d74|C:\\\\WINDOWS\\\\System32\\\\KERNELBASE.dll+c54f6|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncClient.dll+69e811|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\
\\FileSyncClient.dll+f22e8|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncClient.dll+f28ad|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncClient.dll+f1f99|
C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncEvents.dll+f0e0|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncHost.DLL+dc0f|C:\\\\Users\\\\urja\\\\AppData\\
\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncHost.DLL+10485|C:\\\\WINDOWS\\\\System32\\\\ucrtbase.dll+37b0|C:\\\\WINDOWS\\\\System32\\\\KERNEL32.DLL+2e8d7|C:\\\\WINDOWS\\\\SYSTEM32\\\\ntdll.dll+3c34c","sourceUser":"DESKTOP-437I1E6\\\\u
rja","targetUser":"DESKTOP-437I1E6\\\\urja"}}}
2025 Nov 10 02:59:26 (DESKTOP-437I1E6) any->EventChannel {"win":{"system":{"providerName":"Microsoft-Windows-Sysmon","providerGuid":"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}","eventID":"10","version":"3","level":"4","task":"10","opcode":"0","keywords":"0x8
0000000000000","systemTime":"2025-11-09T02:50:46.926401SZ","eventRecordID":"39302","processID":"3088","threadID":"3924","channel":"Microsoft-Windows-Sysmon/Operational","computer":"DESKTOP-437I1E6","severityValue":"INFORMATION","message":"\"Process acc
essed:\r\nRuleName: technique_id=T1036,technique_name=Masquerading\r\nUtcTime: 2025-11-09 02:50:46.924\r\nSourceProcessGUID: {fb82638e-d623-698f-6405-00000000700}\r\nSourceProcessId: 8504\r\nSourceThreadId: 9924\r\nSourceImage: C:\\Users\\urja\\AppData\
\Local\\Microsoft\\OneDrive\\OneDrive.exe\r\nTargetProcessGUID: {fb82638e-f1df-698f-6608-00000000700}\r\nTargetProcessId: 9204\r\nTargetImage: C:\\Users\\urja\\Downloads\\mimikatz_trunk\\x64\\mimikatz.exe\r\nGrantedAccess: 0x101411\r\nCallTrace: C:\\WIN
DOWS\\SYSTEM32\\ntdll.dll+162d74|C:\\WINDOWS\\System32\\KERNELBASE.dll+c54f6|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncClient.dll+69e811|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyn
cClient.dll+f22e8|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncClient.dll+f28ad|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncClient.dll+f1f99|C:\\Users\\urja\\AppData\\Local\\Microsoft
\\OneDrive\\25.194.1005.0003\\FileSyncEvents.dll+f0e0|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncHost.DLL+dc0f|C:\\Users\\urja\\AppData\\Local\\Microsoft\\OneDrive\\25.194.1005.0003\\FileSyncHost.DLL+10485|C:\\WINDOWS
\\System32\\ucrtbase.dll+37b0|C:\\WINDOWS\\System32\\KERNEL32.DLL+2e8d7|C:\\WINDOWS\\SYSTEM32\\ntdll.dll+3c34c\r\nSourceUser: DESKTOP-437I1E6\\urja\r\nTargetUser: DESKTOP-437I1E6\\urja\"}","eventdata":{"ruleName":"technique_id=T1036,technique_name=Masque
rading","utcTime":"2025-11-09 02:50:46.924","sourceProcessGUID":{fb82638e-d623-698f-6405-00000000700}","sourceProcessId":"8504","sourceThreadId":"9924","sourceImage":"C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\OneDrive.exe","ta
rgetProcessGUID":{fb82638e-f1df-698f-6608-00000000700}","targetProcessId":"9204","targetImage":"C:\\\\Users\\\\urja\\\\Downloads\\\\mimikatz_trunk\\\\x64\\\\mimikatz.exe","grantedAccess":"0x101411","callTrace":"C:\\\\WINDOWS\\\\SYSTEM32\\\\ntdll.dll+16
2d74|C:\\\\WINDOWS\\\\System32\\\\KERNELBASE.dll+c54f6|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncClient.dll+69e811|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\
\\FileSyncClient.dll+f22e8|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncClient.dll+f28ad|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncClient.dll+f1f99|
C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncEvents.dll+f0e0|C:\\\\Users\\\\urja\\\\AppData\\\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncHost.DLL+dc0f|C:\\\\Users\\\\urja\\\\AppData\\
\\Local\\\\Microsoft\\\\OneDrive\\\\25.194.1005.0003\\\\FileSyncHost.DLL+10485|C:\\\\WINDOWS\\\\System32\\\\ucrtbase.dll+37b0|C:\\\\WINDOWS\\\\System32\\\\KERNEL32.DLL+2e8d7|C:\\\\WINDOWS\\\\SYSTEM32\\\\ntdll.dll+3c34c","sourceUser":"DESKTOP-437I1E6\\\\u
rja","targetUser":"DESKTOP-437I1E6\\\\urja"}}}
2025 Nov 10 03:00:02 Wazuh->journald Nov 10 03:00:00 Wazuh opensearch-dashboards[76498]: {"type":"log","@timestamp":"2025-11-10T03:00:00Z","tags":["info","plugins","wazuh","monitoring"],"pid":76498,"message":"Settings added to wazuh-monitoring-2025.46w i
ndex"}
2025 Nov 10 03:00:02 Wazuh->journald Nov 10 03:00:00 Wazuh opensearch-dashboards[76498]: {"type":"log","@timestamp":"2025-11-10T03:00:00Z","tags":["info","plugins","wazuh","monitoring"],"pid":76498,"message":"Bulk data to index wazuh-monitoring-2025.46w
for 1 agents completed"}
2025 Nov 10 03:00:04 (DESKTOP-437I1E6) any->EventChannel {"win":{"system":{"providerName":"Microsoft-Windows-Sysmon","providerGuid":"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}","eventID":"3","version":"5","level":"4","task":"3","opcode":"0","keywords":"0x800
0000000000000","systemTime":"2025-11-09T02:51:24.949229Z","eventRecordID":"39303","processID":"3088","threadID":"3932","channel":"Microsoft-Windows-Sysmon/Operational","computer":"DESKTOP-437I1E6","severityValue":"INFORMATION","message":"\"Network conne
ction detected:\r\nRuleName: technique_id=T1036,technique_name=Masquerading\r\nUtcTime: 2025-09-22 12:14:39.145\r\nProcessGuid: {fb82638e-e2e9-68d0-4100-00000000700}\r\nProcessId: 1388\r\nImage: C:\\ProgramData\\Microsoft\\Windows Defender\\Platform\\4.
18.25080.5-0\\MpDefenderCoreService.exe\r\nUser: NT AUTHORITY\\SYSTEM\r\nProtocol: tcp\r\nInitiated: true\r\nSourceIsIpv6: false\r\nSourceIp: 10.0.2.15\r\nSourceHostname: -\r\nSourcePort: 63934\r\nSourcePortName: -\r\nDestinationIsIpv6: false\r\nDestinat
ionIp: 52.123.129.14\r\nDestinationHostname: -\r\nDestinationPort: 443\r\nDestinationPortName: -\"","eventdata":{"ruleName":"technique_id=T1036,technique_name=Masquerading","utcTime":"2025-09-22 12:14:39.145","processGuid":"{fb82638e-e2e9-68d0-4100-0000
00000700}","processId":"1388","image":"C:\\\\ProgramData\\\\Windows Defender\\\\Platform\\\\4.18.25080.5-0\\\\MpDefenderCoreService.exe","user":"NT AUTHORITY\\\\SYSTEM","protocol":"tcp","initiated":"true","sourceIsIpv6":"false","sourceIp":"1
0.0.2.15","sourcePort":"63934","destinationIsIpv6":"false","destinationIp":"52.123.129.14","destinationPort":"443"}}}
2025 Nov 10 03:02:30 Wazuh->df -P ossec: output: 'df -P': Filesystem     1024-blocks     Used Available Capacity Mounted on
2025 Nov 10 03:02:30 Wazuh->df -P ossec: output: 'df -P': tmpfs             812848    1856    811792       1% /run
2025 Nov 10 03:02:30 Wazuh->df -P ossec: output: 'df -P': tmpfs            4064224      80   4064144       1% /dev/shm
2025 Nov 10 03:02:30 Wazuh->df -P ossec: output: 'df -P': tmpfs               5120       0      5120       0% /run/lock
2025 Nov 10 03:02:30 Wazuh->df -P ossec: output: 'df -P': /dev/vda15         106832    6194    100638       6% /boot/efi
2025 Nov 10 03:02:30 Wazuh->df -P ossec: output: 'df -P': tmpfs             812844       4    812840       1% /run/user/0
2025 Nov 10 03:02:30 Wazuh->df -P ossec: output: 'df -P': /dev/vda1       162406320 14359664 148030272       9% /
2025 Nov 10 03:02:30 Wazuh->last -n 20 ossec: output: 'last -n 20':
root     pts/0        107.212.24.97    Mon Nov 10 00:03   still logged in
root     pts/0        107.212.24.97    Sat Nov  8 22:19 - 06:46  (08:27)
root     pts/0        107.212.24.97    Sun Oct 26 20:04 - 02:31  (06:27)
root     pts/0        107.212.24.97    Tue Sep 16 02:38 - 04:50  (02:11)
root     pts/0        107.212.24.97    Tue Sep  9 02:08 - 05:43  (03:34)
root     pts/0        107.212.24.97    Wed Sep  3 16:17 - 16:52  (00:35)
root     tty1                          Wed Sep  3 16:01   still logged in
reboot   system boot  5.15.0-113-gener Wed Sep  3 15:27   still running
wtmp begins Wed Sep  3 15:27:51 2025
2025 Nov 10 03:08:30 Wazuh->df -P ossec: output: 'df -P': Filesystem     1024-blocks     Used Available Capacity Mounted on
2025 Nov 10 03:08:30 Wazuh->df -P ossec: output: 'df -P': tmpfs             812848    1056    811792       1% /run
2025 Nov 10 03:08:30 Wazuh->df -P ossec: output: 'df -P': /dev/vda1       162406320 14359668 148030268       9% /
2025 Nov 10 03:08:30 Wazuh->df -P ossec: output: 'df -P': tmpfs               5120       0      5120       0% /run/lock
2025 Nov 10 03:08:30 Wazuh->df -P ossec: output: 'df -P': /dev/vda15         106832    6194    100638       6% /boot/efi
2025 Nov 10 03:08:30 Wazuh->df -P ossec: output: 'df -P': tmpfs             812844       4    812840       1% /run/user/0
2025 Nov 10 03:08:30 Wazuh->df -P ossec: output: 'df -P': tmpfs            4064224      80   4064144       1% /dev/shm
2025 Nov 10 03:08:30 Wazuh->last -n 20 ossec: output: 'last -n 20':
root     pts/0        107.212.24.97    Mon Nov 10 00:03   still logged in
root     pts/0        107.212.24.97    Sat Nov  8 22:19 - 06:46  (08:27)
root     pts/0        107.212.24.97    Sun Oct 26 20:04 - 02:31  (06:27)
root     pts/0        107.212.24.97    Tue Sep 16 02:38 - 04:50  (02:11)
root     pts/0        107.212.24.97    Tue Sep  9 02:08 - 05:43  (03:34)
root     pts/0        107.212.24.97    Wed Sep  3 16:17 - 16:52  (00:35)
root     tty1                          Wed Sep  3 16:01   still logged in
reboot   system boot  5.15.0-113-gener Wed Sep  3 15:27   still running
wtmp begins Wed Sep  3 15:27:51 2025
2025 Nov 10 03:10:02 Wazuh->journald Nov 10 03:10:01 Wazuh CRON[721706]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
2025 Nov 10 03:10:02 Wazuh->journald Nov 10 03:10:01 Wazuh CRON[721707]: (root) CMD (test -e /run/systemd/system || SERVICE_MODE=1 /sbin/e2scrub_all -A -r)
2025 Nov 10 03:10:02 Wazuh->journald Nov 10 03:10:01 Wazuh CRON[721706]: pam_unix(cron:session): session closed for user root
root@Wazuh:/var/ossec/logs/archives#
```

Description: Modifying the Filebeat configuration by enabling archive and alerts to true and restarting the Filebeat service.

```
  GNU nano 6.2
# Wazuh - Filebeat configuration file
output.elasticsearch.hosts:
        - 127.0.0.1:9200
#        - <elasticsearch_ip_node_2>:9200
#        - <elasticsearch_ip_node_3>:9200

output.elasticsearch:
  protocol: https
  username: ${username}
  password: ${password}
  ssl.certificate_authorities:
    - /etc/filebeat/certs/root-ca.pem
  ssl.certificate: "/etc/filebeat/certs/wazuh-server.pem"
  ssl.key: "/etc/filebeat/certs/wazuh-server-key.pem"
setup.template.json.enabled: true
setup.template.json.path: '/etc/filebeat/wazuh-template.json'
setup.template.json.name: 'wazuh'
setup.ilm.overwrite: true
setup.ilm.enabled: false

filebeat.modules:
  - module: wazuh
    alerts:
      enabled: true
    archives:
      enabled: true

logging.level: info
logging.to_files: true
logging.files:
  path: /var/log/filebeat
  name: filebeat
  keepfiles: 7
  permissions: 0644

logging.metrics.enabled: false

seccomp:
  default_action: allow
  syscalls:
  - action: allow
    names:
    - rseq
```

Description: Defining an index pattern in Dashboard Management for indexing and searching logs.



Create index pattern - Wazuh

Not Secure https://104.131.168.175/app/management/opensearch-dashboards/indexPatterns/create

Dashboards Ma... | Index patterns | Create index pattern

Dashboards Management

- Index patterns
- Data sources
- Saved objects
- Advanced settings

Create index pattern

An index pattern can match a single source, for example, `filebeat-4-3-22` , or **multiple** data sources, `filebeat-*` .

Read documentation

Step 2 of 2: Configure settings

Specify settings for your **wazuh-archives-*** index pattern.

Select a primary time field for use with the global time filter.

Time field | Refresh

timestamp

Show advanced settings

Back | Create index pattern

Discover - Wazuh

Not Secure https://104.131.168.175/app/data-explorer/discover#?_a=(discover:(columns:!(_source),isDirty:!f,sort:!()),metadata:(indexPattern:bcaf54f0-c94c-11f0-ae8e-5928dfab88a7,view:di...

Discover | New | Save | Open | Share | Reporting | Inspect

wazuh-archives-*

Search field names | Filter by type 0

Selected fields

- _source

Available fields

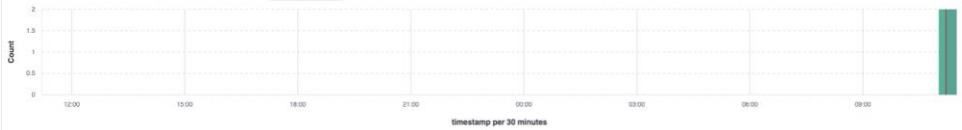
- _index
- @timestamp
- agent.id
- agent.ip
- agent.name
- data.win.eventdata.callTrace
- data.win.eventdata.commandLine
- data.win.eventdata.company
- data.win.eventdata.creationUtcTime
- data.win.eventdata.currentDirectory
- data.win.eventdata.description
- data.win.eventdata.destinationIp
- data.win.eventdata.destinationIsIpv6
- data.win.eventdata.destinationPort
- data.win.eventdata.details
- data.win.eventdata.eventType
- data.win.eventdata.fileVersion

mimikatz | DQL | Last 24 hours | Show dates | Refresh

Add filter

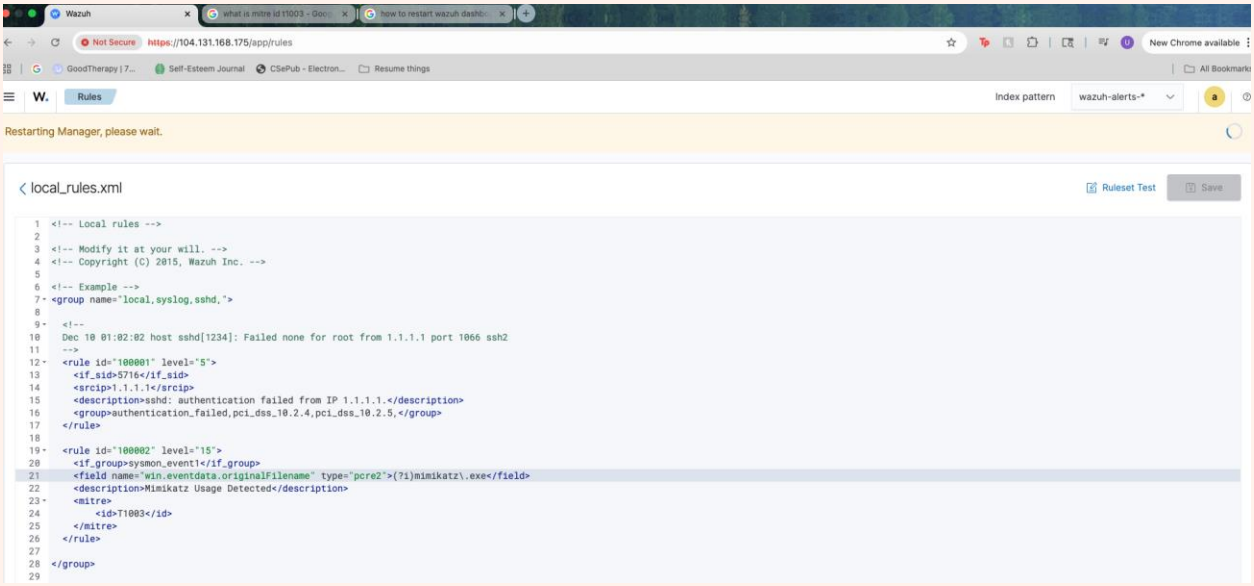
2 hits

Nov 23, 2025 @ 11:12:16.121 - Nov 24, 2025 @ 11:12:16.121 per Auto

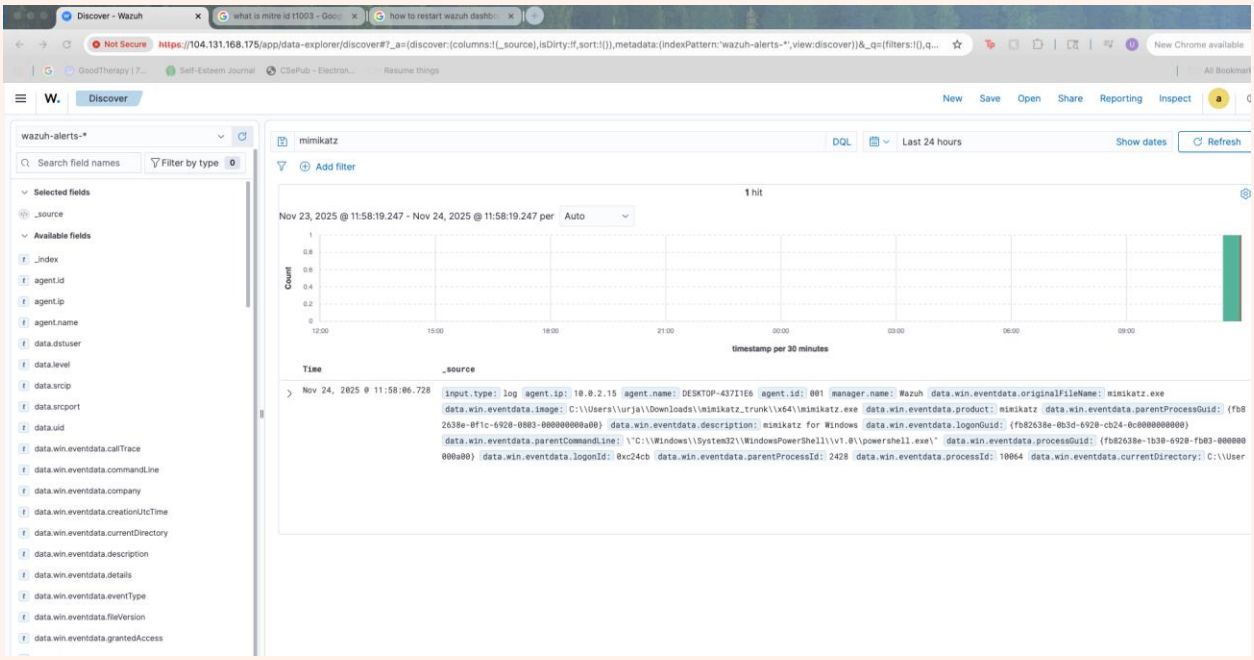


Time	_source
Nov 24, 2025 @ 11:11:55.406	agent.ip: 10.0.2.15 agent.name: DESKTOP-437I1E6 agent.id: 001 manager.name: Wazuh data.win.eventdata.originalFileName: mimikatz.exe data.win.eventdata.image: C:\\Users\\urja\\Downloads\\mimikatz_trunk\\x64\\mimikatz.exe data.win.eventdata.product: mimikatz data.win.eventdata.imageLoaded: C:\\Users\\urja\\Downloads\\mimikatz_trunk\\x64\\mimikatz.exe data.win.eventdata.description: mimikatz for Windows data.win.eventdata.signed: false data.win.eventdata.signatureStatus: Unavailable data.win.eventdata.processGuid: {fb82638e-1712-6920-9703-000000000a00} data.win.eventdata.processId: 912 data.win.eventdata.utcTime: 2025-11-21 07:38:58.549 data.win.eventdata.hashes: SHA1=E3B6EA8C46FA831CEC6F235A5CF48B38A4AE8D69,MD5=29EFD64DD3C7FE1E2B022B7AD73A1BA5,SHA256=61C0810A23580CF492A6BA4F76545661088331E7A4134C968C2D6
Nov 24, 2025 @ 11:11:55.400	agent.ip: 10.0.2.15 agent.name: DESKTOP-437I1E6 agent.id: 001 manager.name: Wazuh data.win.eventdata.originalFileName: mimikatz.exe data.win.eventdata.image: C:\\Users\\urja\\Downloads\\mimikatz_trunk\\x64\\mimikatz.exe data.win.eventdata.product: mimikatz data.win.eventdata.parentProcessGuid: {fb82638e-0f1c-6920-0803-000000000a00} data.win.eventdata.description: mimikatz for Windows data.win.eventdata.logonGuid: {fb82638e-0b3d-6920-cb24-0c0000000000} data.win.eventdata.parentCommandLine: \"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\" data.win.eventdata.processGuid: {fb82638e-1712-6920-9703-000000000a00} data.win.eventdata.logonId: 0xc24cb data.win.eventdata.parentProcessId: 2428 data.win.eventdata.processId: 912 data.win.eventdata.currentDirectory: C:\\Users\\urja\\Downloads\\mimikatz_trunk\\x64\\

Description: Editing local rules to add a detection for Mimikatz. Triggering the rule when the original file name for Mimikatz appears in Sysmon Event ID 1 (process creation). Mapping the detection to the MITRE ATT&CK framework under T1003 – OS Credential Dumping.



Description: Restarting the Wazuh dashboard and executing Mimikatz again on the Windows machine. Using the Discover function to verify that the Mimikatz detection rule is generated.



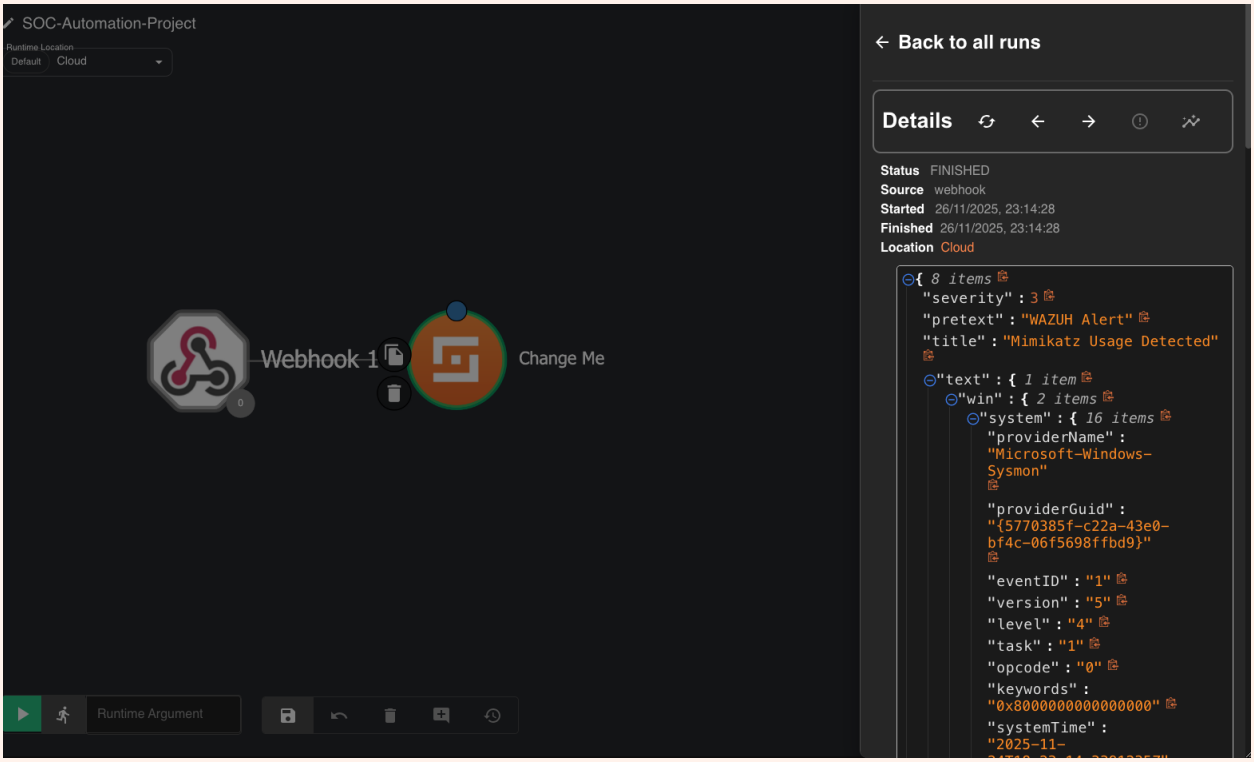
t id	1764003486.722142
t input.type	log
t location	EventChannel
t manager.name	Wazuh
t rule.description	Mimikatz Usage Detected
# rule.firedtimes	1
t rule.groups	local, syslog, sshd
t rule.id	100002
# rule.level	15
rule.mail	true
t rule.mitre.id	T1003
t rule.mitre.tactic	Credential Access
t rule.mitre.technique	OS Credential Dumping
timestamp	Nov 24, 2025 @ 11:58:06.728

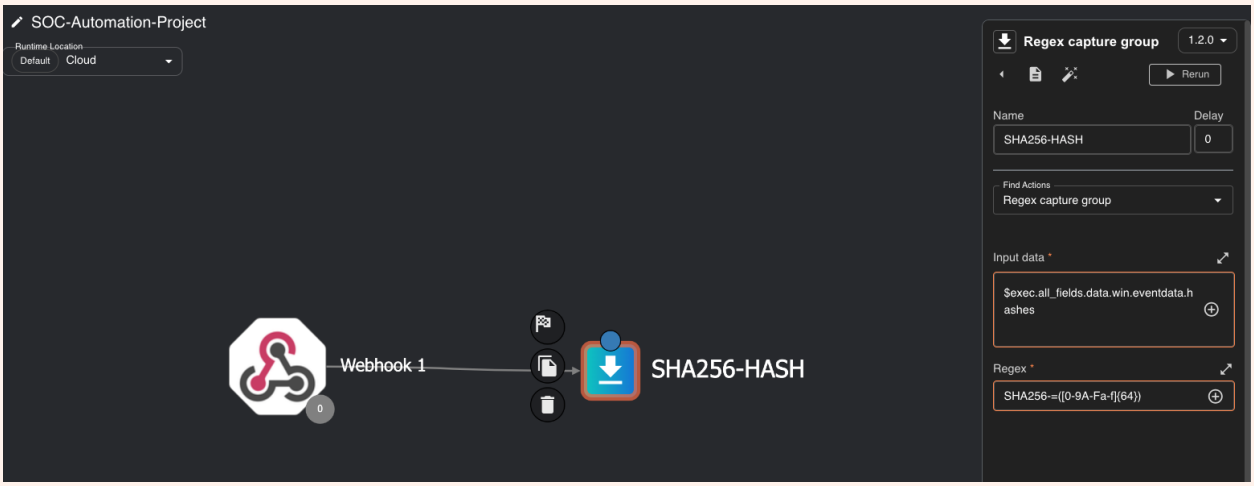
Description: Adding the Shuffle integration to the OSSEC configuration file and restarting the Wazuh manager.

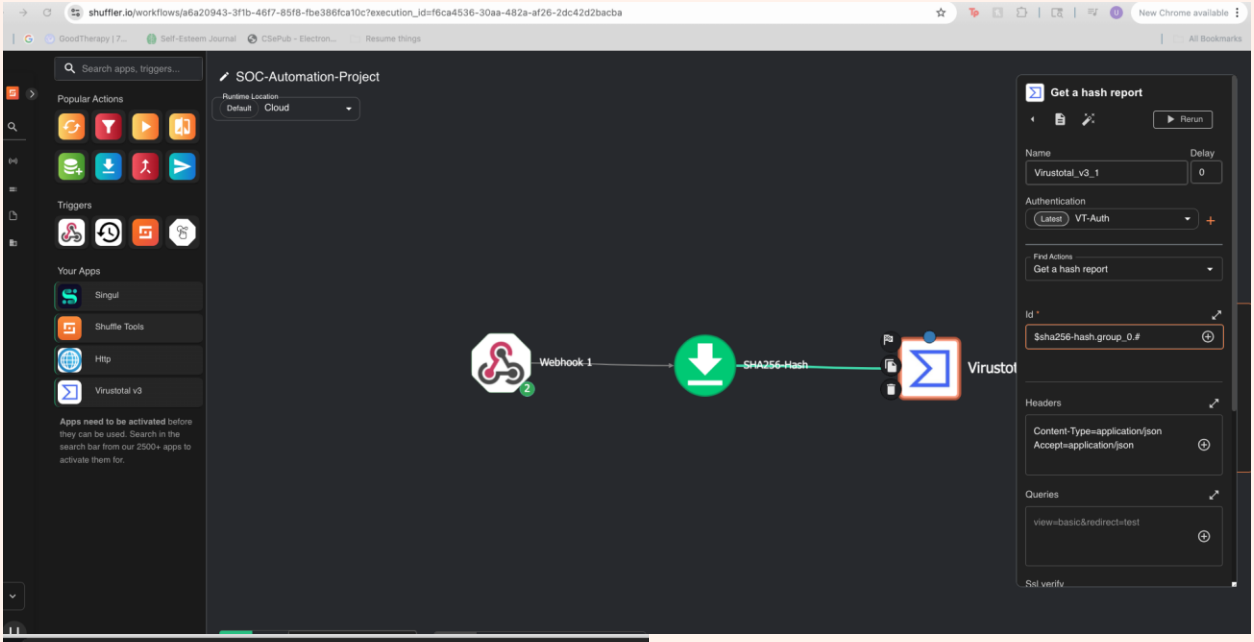
```
--
 Wazuh - Manager - Default configuration for ubuntu 22.04
 More info at: https://documentation.wazuh.com
 Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

ossec_config>
 <global>
   <jsonout_output>yes</jsonout_output>
   <alerts_log>yes</alerts_log>
   <logall>yes</logall>
   <logall_json>yes</logall_json>
   <email_notification>no</email_notification>
   <smtp_server>smtp.example.wazuh.com</smtp_server>
   <email_from>wazuh@example.wazuh.com</email_from>
   <email_to>recipient@example.wazuh.com</email_to>
   <email_maxperhour>12</email_maxperhour>
   <email_log_source>alerts.log</email_log_source>
   <agents_disconnection_time>10m</agents_disconnection_time>
   <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
   <update_check>yes</update_check>
 </global>

 <integration>
   <name>shuffle</name>
   <hook_url>https://shuffler.io/api/v1/hooks/webhook_48f9259c-160e-4a33-80fa-9abd6b1e3dba </hook_url>
   <rule_id>100002</rule_id>
   <alert_format>json</alert_format>
 </integration>


 <alerts>
   <log_alert_level>3</log_alert_level>
   <email_alert_level>12</email_alert_level>
 </alerts>

 <!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
 <logging>
   <log_format>plain</log_format>
 </logging>

 <remote>
   <connection>secure</connection>
   <port>1514</port>
   <protocol>tcp</protocol>
   <queue_size>131072</queue_size>
 </remote>
```

Description: Build workflow in shuffle soar using webhooks:

1. Mimikatz Alert Sent to Shuffle
2. Shuffle Receives Mimikatz Alert – Extracting SHA256 Hash From File
3. Check Reputation Score with Virus total
4. Send details to Hive to create alert
5. Send email to SOC Analyst to begin Investigation







← Back to all runs

Details

Status FINISHED
Source webhook
Started 27/11/2025, 00:03:48
Finished 27/11/2025, 00:03:49
Location Cloud


```
{ 8 items
  "severity" : 3
  "pretext" : "WAZUH Alert"
  "title" : "Mimikatz Usage Detected"
  "text" : { 1 item
    "win" : { 2 items
      "system" : {...} 16 items
      "eventdata" : {...}
       23 items
    }
  }
  "rule_id" : "100002"
  "timestamp" :
  "2025-11-27T05:03:45.323+0000"
  "id" : "1764219825.154145"
  "all_fields" : {...} 9 items
}
```


SHA256-Hash
regex_capture_group

← Back to all runs

Details

Status FINISHED
Source webhook
Started 27/11/2025, 00:03:48
Finished 27/11/2025, 00:03:49
Location Cloud


```
{ 8 items
  "severity" : 3
  "pretext" : "WAZUH Alert"
  "title" : "Mimikatz Usage Detected"
  "text" : { 1 item
    "win" : { 2 items
      "system" : {...} 16 items
      "eventdata" : {...}
       23 items
    }
  }
  "rule_id" : "100002"
  "timestamp" :
  "2025-11-27T05:03:45.323+0000"
  "id" : "1764219825.154145"
  "all_fields" : {...} 9 items
}
```

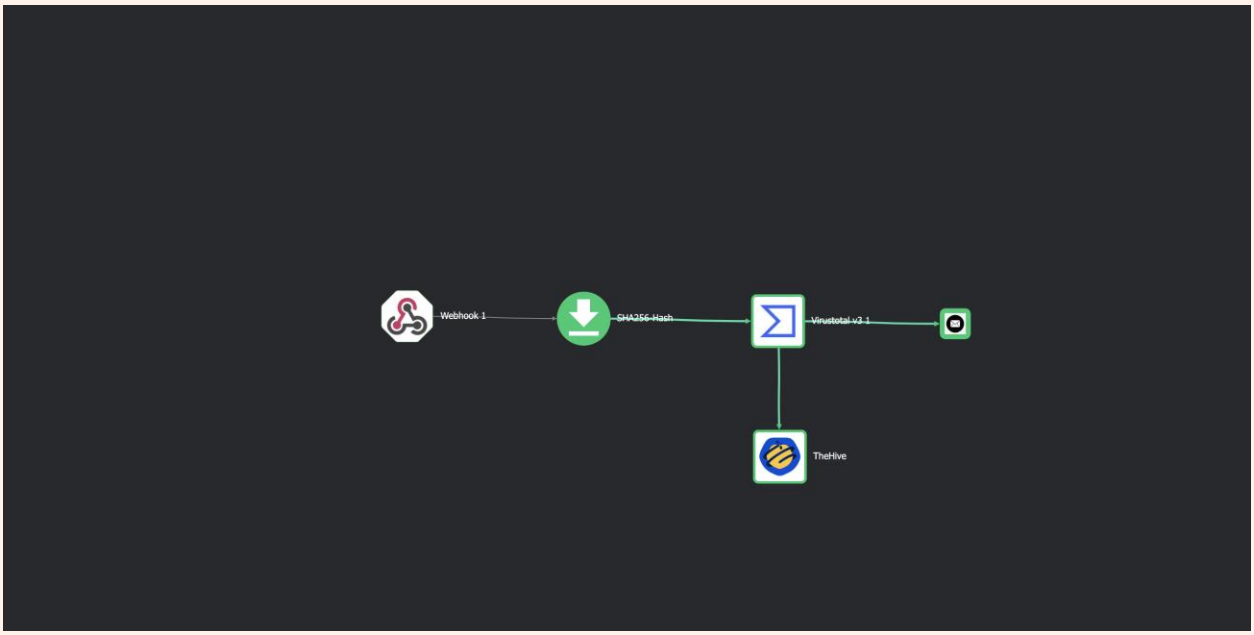

SHA256-Hash
regex_capture_group

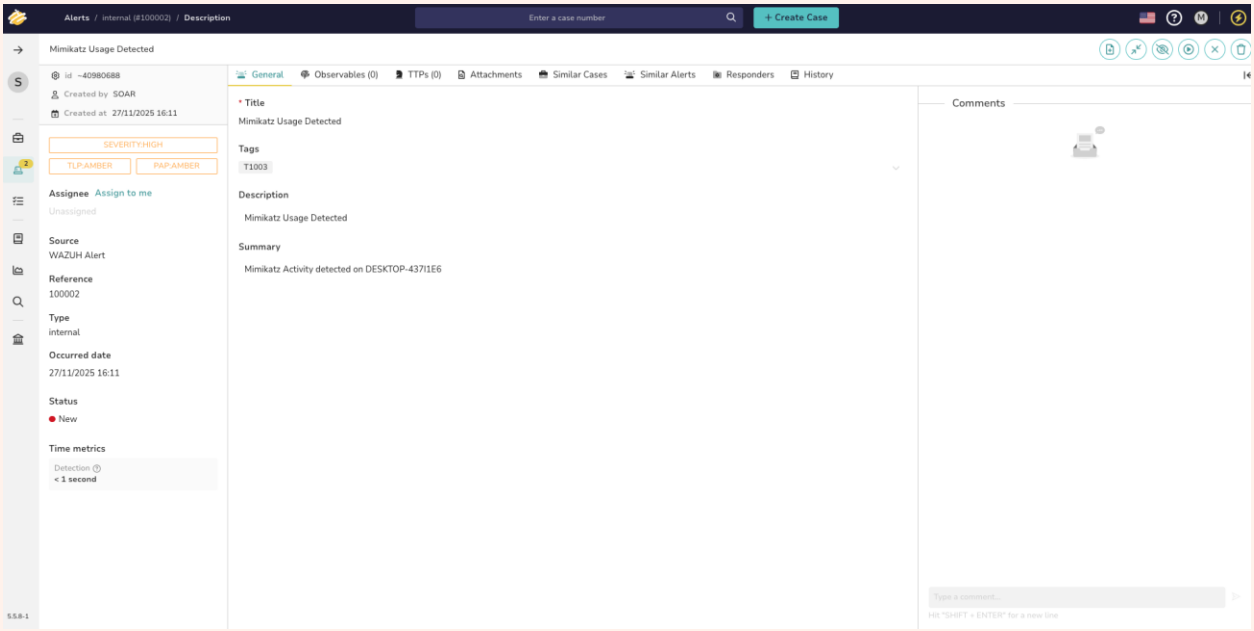
SHA256-Hash
regex_capture_group

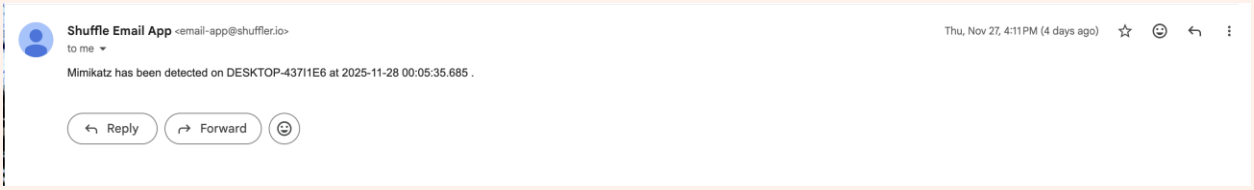

```
"Results for SHA256-Hash" : {
 3 items
  "success" : true
  "group_0" : [...] 1 item
  "found" : true
}
```


Virustotal v3 1
get_a_hash_report_


```
"Results for Virustotal_v3_1" : [
 1 item
  0 : { 6 items
    "status" : 200
    "body" : { 1 item
      "data" : { 4 items
        "id" :
        "61c0810a23580cf492a6ba4f7654
        "type" : "file"
        "links" : { 1 item
          "self" :
          "https://www.virustotal.com
          ..."
        }
        "attributes" : {...}
         41 items
```


Webhook 1
SHA256 Hash
Virustotal v3 1
TheHive


Alerts / internal (#100002) / Description
Enter a case number
+ Create Case
Mimikatz Usage Detected
id ~40980688
Created by SOAR
Created at 27/11/2025 16:11
SEVERITY:HIGH
TLP:AMBER
PAP:AMBER
Assignee Assign to me
Unassigned
Source
WAZUH Alert
Reference
100002
Type
internal
Occurred date
27/11/2025 16:11
Status
New
Time metrics
Detection
< 1 second
General
Observables (0)
TTPs (0)
Attachments
Similar Cases
Similar Alerts
Responders
History
* Title
Mimikatz Usage Detected
Tags
T1003
Description
Mimikatz Usage Detected
Summary
Mimikatz Activity detected on DESKTOP-437I1E6
Comments
Type a comment...
Hit "SHIFT + ENTER" for a new line
5.5.8-1


Shuffle Email App <email-app@shuffler.io>
to me
Thu, Nov 27, 4:11 PM (4 days ago)
Mimikatz has been detected on DESKTOP-437I1E6 at 2025-11-28 00:05:35.685 .
Reply
Forward